

RESEARCH ARTICLE

Development of a composite using calcined bone powder and silane cross-linked alginate as bone substitute material

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Abstract

Calcined bone is an attractive natural material for use as a bone substitute because of its cost-effectiveness and high biocompatibility, which are comparable to that of synthetic hydroxyapatite. However, the calcination process has significantly weakened the mechanical properties. In this study, a composite of calcined bovine bone powder reinforced with silane cross-linked alginate was prepared to assess its biocompatibility, osteoconductivity, and mechanical compatibility as a bone substitute material. Culture studies with osteoblast-like cells (MC3T3-E1) showed no cytotoxicity toward the composite and exhibited general cell proliferative properties in its presence. In contrast, the composite reduced the alkaline phosphatase activity of osteoblasts but led to significant noncellular apatite deposition on the surface. In addition, quasi-static compression tests of the composite revealed mechanical properties comparable to those of human cancellous bone. The mechanical properties remained stable under wet conditions and did not deteriorate significantly even after 2 weeks of immersion in simulated body fluid at 37°C. The results show that this composite, composed of calcined bone powder and silane cross-linked alginate, is a promising bone substitute material with biocompatibility, osteoconductivity, and mechanical compatibility.

KEYWORDS

alginate, bone substitute material, calcined bone, composite, silane cross-linking

1 | INTRODUCTION

Osteoporosis is a bone disease in which bone strength decreases. This makes bone fractures more likely as bone mass decreases and the microstructure of the bone deteriorates. The number of osteoporotic fractures worldwide is estimated at 9 million per year.¹ Osteoporosis-related fractures can occur even after minor falls in everyday life, and can significantly affect the quality of life of older people.² Given the diminished natural healing capacity of the elderly, the treatment of fractures has mainly been performed by grafting artificial bone made

from various industrial materials, such as titanium alloys,³ hydroxyapatite,⁴ and ultra-high molecular weight polyethylene.⁵ One advantage of artificial bone transplantation is that it is less stressful for the patient as, unlike autologous bone transplantation there is no invasion of the grafting site. However, one drawback is that it does not replace the autologous bone, which in some cases leads to problems such as extraction and reimplantation after the initial implantation.⁶ An ideal artificial bone requires not only biocompatibility but also bioabsorbability, and osteoinductive and osteoconductive capabilities for integration into the bone in vivo. It should also have the ability to be

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resorbed and replaced by the patient's bone tissue after implantation. In addition, initial rigidity and mechanical compatibility are essential elements to avoid stress shielding. However, in today's artificial bones, there is no single material that fulfills these requirements.

Research and development is underway to utilize natural organic and inorganic materials as biomaterials, and efforts are being made to use them for the development of artificial bones. Collagen has excellent cell adhesion properties and is widely used as a natural polymeric cell scaffold material.⁷ In addition, polysaccharides such as chitin⁸ extracted from crustaceans, plant-derived starch,⁹ hyaluronic acid, which is widely distributed in mammalian connective tissues,¹⁰ and alginate obtained from brown algae, such as kelp,¹¹ have also been used as biodegradable scaffold materials owing to their biocompatibility. However, these materials are relatively soft and may not offer the initial rigidity required after grafting into bone, which poses challenge for their use as artificial bones. Alternatively, natural inorganic materials such as coral-derived calcium carbonate have been investigated as scaffold materials.¹² Biological bone apatite is another remarkable material. Unlike artificially synthesized hydroxyapatite (HAP), it is known to contain carbonate groups and is referred to as carbonate apatite.¹³ Carbonate apatite has a higher solubility than HAP, making it a promising material for use as a bone substitute with excellent biodegradability and tissue replacement capabilities. However, these inorganic materials are generally brittle and exhibit poor mechanical compatibility, which presents a significant challenge in their application as bone substitute materials. Therefore, various studies have been conducted to address this brittleness issue by integrating it with biocompatible organic materials such as polycaprolactone¹⁴ and poly-D-L-lactide.¹⁵ However, these efforts have not produced the necessary toughness to function as an effective bone substitute material.

The use of xenografts from animal bone as bone substitute material is not the first choice due to problems such as immunogenicity, infection risk, and low engraftment rates.¹⁶ In contrast to the autogenous bone, however, it has the advantage of providing a stable supply. To improve the practicality of the xenogeneic bone graft, studies have investigated decellularization to remove immunogenicity,¹⁷ and various sterilization methods such as gamma radiation,¹⁸ chemical treatment, calcination,¹⁹ and the application of supercritical CO₂ fluid²⁰ to eliminate infectivity. Calcination is an excellent method for effectively eliminating organic components, such as proteins, from heterologous bone, effectively removing immunogenicity, and infectivity. In the field of dental treatment for example, products such as Bio-Oss® (Geistlich Pharma, Wolhusen, Switzerland), which are produced by high-temperature calcination of bovine bones, have already been successfully commercialized. However, the loss of collagen fibers through calcination results in low mechanical toughness, which makes load-bearing applications difficult. Consequently, it is primarily used as a bone graft material for alveolar and jawbone applications.

Therefore, in this study, a novel bone substitute material was developed by combining calcined bovine bone particles with alginate, a natural polysaccharide extracted from seaweed and a cost-effective

biomaterial. It has been widely studied in the fields of regenerative medicine and drug delivery systems, owing to its high biocompatibility²¹ and ease of chemical modification.²² Sodium alginate is a neutral salt consisting of a carboxyl group bonded to a Na ion. When dissolved in water, it becomes a viscous aqueous solution. The addition of multivalent metal ions, such as Ca and Al ions, results in cross-linking of the carboxyl groups of alginic acid and metal ions to form a gel. In this study, a composite was proposed as a new bone substitute material, consisting of calcined bone particles that contribute to mechanical rigidity and alginate fibers, which provide fracture resistance. This composite is expected to have high mechanical toughness and mechanical compatibility. However, there is a problem with alginate gel when placed in body fluids, as swelling of the gel occurs by leaching calcium ions from the gel.²³ Consequently, the mechanical properties of the composite deteriorated considerably soon after transplantation.

Covalent cross-linking of alginate fibers occurs through 3-aminopropyltriethoxysilane promoted silane coupling, and alginate spontaneously forms apatite by incorporating calcium and phosphate ions from body fluids.²⁴ Therefore, it was hypothesized that the calcined bone powder/alginate composite could obtain the necessary mechanical rigidity, mechanical stability in bodily fluids, and osteoconductivity through silane coupling. The objective of this study was to validate this hypothesis by preparing a composite, examining its mechanical properties under quasi-static loading, and observing the changes in simulated body fluids (SBF). In addition, the cytotoxicity of the composite using cultured osteoblast cells was assessed and the impact of the composite on cell proliferation, differentiation, and bone matrix formation was investigated.

2 | MATERIALS AND METHODS

2.1 | Calcined bone powder

First, a frozen bovine femur was cut into the distal trabecular bone ends, boiled for 30 min to remove excess bone marrow components, dried naturally, and then calcined at 600°C for 22 h in an electric furnace to remove organic components. The calcined trabecular bone block was ground in a mortar and sieved through a sieve with a 500 µm aperture. The particle size of the calcined bone powder was measured using a Laser Scattering Particle Size Distribution Analyzer (HORIBA, LA-960, Kyoto, Japan), and the crystal structure was analyzed using an x-ray diffractometer (XRD) (BRUKER, D2 PHASER, MA, United States). The chemical structure of the calcined bone powder was analyzed by Fourier-transform infrared spectrometry (FTIR) (HORIBA, FT-730, Kyoto, Japan) using the KBr method.

2.2 | Silane cross-linked alginate

Silane cross-linked alginate was prepared according to the method described by Hosoya et al.²⁴ Briefly, 3-aminopropyltriethoxysilane

(APES, 1.5×10^{-2} mol/L, 301-60441, FUJIFILM Wako Pure Chemical Corp., Osaka, Japan), a silane coupling agent, 1-ethyl-3-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC·HCl, 0.98×10^{-1} mol/L, D1601, Tokyo Chemical Industry Co., Tokyo, Japan) and N-hydroxysuccinimide (HOSu, 1.5×10^{-2} mol/L, 089-0403, FUJIFILM Wako Pure Chemical Corp., Osaka, Japan), which promote the dehydration condensation of the carboxyl groups in alginic acid and the amino groups in the reactants, were added to a 1% sodium alginate solution (199-09961, FUJIFILM Wako Pure Chemical Corp., Osaka, Japan) and mixed. The mixture was poured into an acrylic mold and allowed to gel at room temperature for 3 days. To remove unreacted reagents and the resulting products, the gel was washed with a sodium chloride aqueous solution (143×10^{-3} mol/L), an aqueous calcium chloride solution (2.5×10^{-3} mol/L), and pure water. They were then cut into cubic blocks using a surgical scalpel and freeze-dried. Finally, the dried gel was soaked in a calcium chloride solution (1 mol/L) at 36.5°C for 24 h, and then rinsed with pure water at room temperature for 30 min. To confirm the apatite formation ability of the silane cross-linked alginate prepared as described above, it was immersed in SBF²⁵ for 1 or 2 weeks. After immersion, the sample was freeze-dried, and the observation surface was subjected to Au-Pd sputtering (E-1030, HITACHI, Tokyo, Japan). The morphology of the deposit on the sample surface was observed using field-emission scanning electron microscopy (FE-SEM, HITACHI, JSM-7100F, Tokyo, Japan) and elemental analysis was performed using energy-dispersive x-ray analysis (EDX). The crystal structure was analyzed using an x-ray diffractometer (XRD, Rigaku, Mini Flex 600, Tokyo, Japan).

2.3 | Calcined bone powder/silane cross-linked alginate composite

During the silane cross-linked alginate preparation process described above, calcined bone powder was added to the mixture of sodium alginate, APES, EDC·HCl, and HOSu at a mass ratio of 1:1. The mixture was then placed in an aluminum mold with a diameter of 15 mm and a height of 5 mm and allowed to gel for 3 days, and then washed. Finally, the calcined bone powder/silane cross-linked alginate composite (B/A-Si) was obtained.

2.4 | Mechanical properties

To investigate the mechanical stability of B/A-Si in body fluids, immersion experiments were performed in SBF for 1 or 2 weeks. Immediately after immersion, a quasi-static compression test was performed using a mechanical testing machine (AG-250kNXPlus, SHIMADZU, Kyoto, Japan) at a displacement rate of 1 mm/min and a maximum displacement of 1 mm. The elastic modulus and the maximum stress were determined from the stress-strain curve obtained.

2.5 | Cytotoxicity

All cell culture experiments were performed using the osteoblast-like cell line, MC3T3-E1 (RCB1126) provided by the RIKEN BRC as part of the National Bio-Resource Project of the MEXT/AMED, Japan.

The cytotoxicity of B/A-Si was evaluated by extraction of the composite. B/A-Si was placed in a 50 mL centrifuge tube, and α -modified minimum essential medium (M0894, Sigma-Aldrich, MO, United States) supplemented with 10% fetal bovine serum and 1% antibiotics (100 U/mL penicillin and 0.1 mg/mL streptomycin; 15,140-122, Thermo Fisher Scientific, MA, United States) was added at a ratio of 10 mL per 1 g of sample. It was incubated for 24 h at 37°C with 95% humidity and 5% CO_2 and used as the extraction solution for B/A-Si. The extract was mixed with fresh medium at ratios of 0%, 25%, 50%, and 100% to assess alterations in osteoblast viability associated with varying proportions of the extract. MC3T3-E1 cells were seeded into 6-well plates at a density of 7.5×10^4 cells per well and cultured in the mixed medium for 4 days. Dead cells were stained with trypan blue and cell counts were determined using a hemocytometer. Cell viability was calculated as the percentage of viable cells in relation to the total number of cells.

2.6 | Cellular proliferation

The effect of B/A-Si on the proliferation of osteoblasts was investigated using a WST-8 assay (Cell Counting Kit-8; Dojindo, Kumamoto, Japan). MC3T3-E1 cells were seeded at a density of 5×10^4 cells/well on B/A-Si placed in a 24-well plate and cultured in the aforementioned medium. On day 0 (6 h after seeding), 2, and 4, reagents were added to the wells according to the manufacturer's instructions, and the absorbance was measured at 450 nm using a spectrophotometer (U-1900, Hitachi, Tokyo, Japan). The absorbance values obtained were expressed as a ratio to the values at day 0 and used to evaluate cell numbers. In addition, cells cultured in 24-well plates were used as controls and the results were compared.

2.7 | Alkaline phosphatase activity

The effect of B/A-Si on the alkaline phosphatase (ALP) activity in osteoblasts was investigated. MC3T3-E1 cells were seeded at 1×10^6 cells/well on B/A-Si placed in a 24-well plate. After 1, 3, and 7 days of culture, the cells were lysed using lysis buffer, frozen, thawed, vortexed, and then centrifuged. The supernatant was used as protein solution. ALP activity was determined by spectrophotometrically measuring the absorbance of p-nitrophenol produced as a degradation product in the protein solution at 405 nm using an ALP assay kit (LabAssay™ ALP, FUJIFILM Wako Chemical Corp. Osaka, Japan) according to the manufacturer's instructions. In addition, the ALP activity of cells cultured without B/A-Si was measured and used as a control.

2.8 | Cellular morphology

Apatite formation and cell morphology on B/A-Si in culture were observed by FE-SEM. In a 6-well plate, 5×10^4 cells were seeded on B/A-Si samples, and after 1 or 2 weeks of cultivation, tissue fixation by mixture of 2% paraformaldehyde and 2% glutaraldehyde in 0.2 M sodium cacodylate buffer, pH 7.4, stepwise dehydration with ethanol. The samples were freeze-dried and then sputter-coated with an Au-Pd alloy on the observation surface. The B/A-Si samples, which were cultured without cell seeding, were observed in the same manner.

2.9 | Osteogenesis

To investigate the formation of new bone matrix by osteoblasts on B/A-Si, osteoblasts were seeded with 1.0×10^6 cells on the composite in a 24-well plate, cultured for 21 days, fixed, dehydrated, and freeze-dried, as described above. Samples were imaged with a hydroxyapatite phantom using a microfocus x-ray CT system (InspeXio SMX-225CT FPD HR; Shimadzu Corporation, Kyoto, Japan). The CT images were analyzed using 3D bone morphometry software (TRI/3D-BON, RATOC System Engineering, Tokyo, Japan) to extract the density below 500 mg/cm^3 as new bone matrix and determine its mass (nBMC, mg), volume (nBV, mm^3), and density (nBMD, mg/cm^3). For comparison, normally cultured cells on a plastic surface (control) and B/A-Si cultured without cells were prepared, and the above three characteristics were determined.

2.10 | Statistics

All experiments were performed with three samples for each condition, and statistical analyses of the data were performed using Kalei-daGraph software version 3.6 (Synergy Software, Reading, PA, United States). In the experiments on mechanical properties, cell viability, and new bone matrix formation, one-way analysis of variance (ANOVA) confirmed the presence of significant differences in all groups, and Tukey's post hoc tests were used to identify significant differences between groups. For cell proliferation and ALP activity experiments, *t*-tests confirmed significant differences compared to controls. In all statistical analyses, a *p*-value of less than .05 was considered statistically significant.

3 | RESULTS

3.1 | Calcined bone powder

The particle size distribution ranged from sub- μm to several hundred micrometers, with a median of $95.5 \mu\text{m}$, an average of $246 \mu\text{m}$, and a mode of $148 \mu\text{m}$ (Figure 1). X-ray diffraction analysis of the calcined bone powder showed a similar spectrum to that of hydroxyapatite

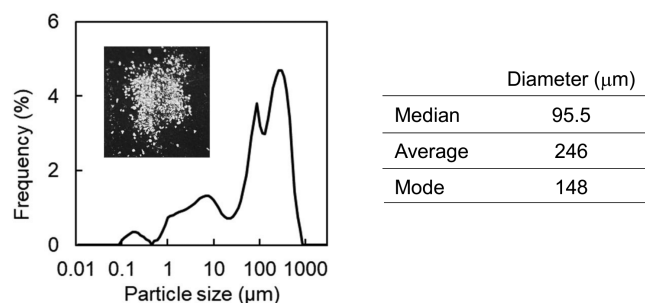


FIGURE 1 Particle size distribution of calcined bone powder. The particles are widely distributed in size, primarily in the range of several hundred micrometers (μm).

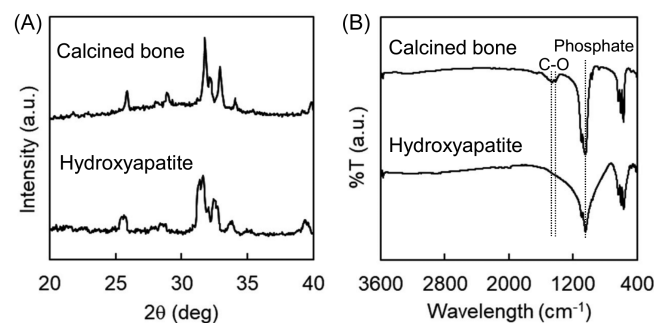


FIGURE 2 XRD (A) and FTIR (B) spectra of calcined bone and hydroxyapatite. Both have a similar crystal structure, but carbonate groups are present in the calcined bone at 600°C . FTIR, Fourier-transform infrared spectrometry; XRD, x-ray diffractometer.

(NEOBONE[®], CoorsTek, Tokyo, Japan) (Figure 2A). The FTIR spectrum of the calcined bone powder was similar to that of hydroxyapatite. However, the presence of peaks at 1410 and 1460 cm^{-1} originating from carbonate groups, indicates that it is carbonate apatite (Figure 2B).

3.2 | Silane cross-linked alginate

The surface of the silane cross-linked alginate was smooth before immersion in SBF. However, immersion led to the formation of surface deposits, the size of which increased with longer immersion time (Figure 3B,C). The XRD analysis revealed that these deposits had a hydroxyapatite-like crystal structure (Figure 4A). With increasing immersion time, peaks at $2\theta = 26^\circ$ and 32° appeared in the XRD spectrum, resembling the characteristic peaks of hydroxyapatite. EDX analysis showed that prior to immersion, in addition to the peaks of the constituent elements of alginate, Si peaks were detected, indicating the occurrence of silane coupling in the sample (Figure 4B). One week after immersion, a phosphate peak was detected, indicating apatite deposition (Figure 4C). The peak became clearer 2 weeks after immersion, showing the progression of apatite deposition (Figure 4D).

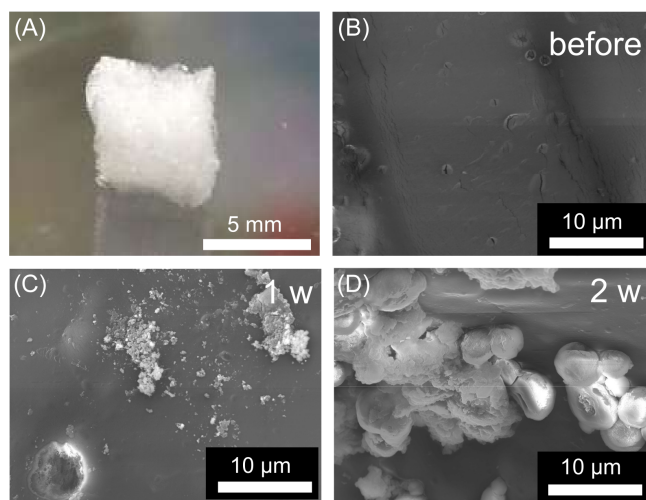


FIGURE 3 Close-up images of a silane cross-linked alginate sample (A) and FE-SEM images of the surfaces before (B), 1 week (C), and 2 weeks (D) after immersion in SBF. Apatite-like deposits formed on the surface of the silane cross-linked alginate with increasing immersion time. FE-SEM, field-emission scanning electron microscopy; SBF, simulated body fluid.

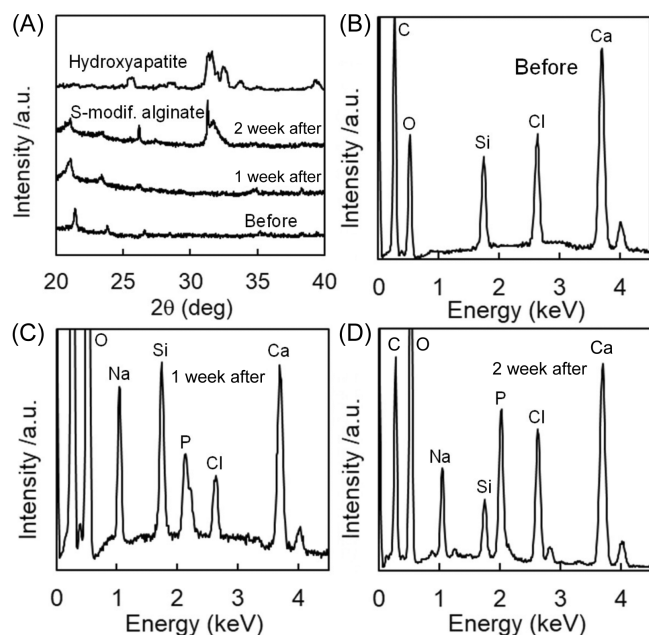


FIGURE 4 XRD (A) and EDX spectra of silane cross-linked alginate before (B), 1 week after (C), and 2 weeks after (D) immersion in SBF. Surface deposits of silane cross-linked alginate exhibit a hydroxyapatite-like crystal structure. The increase in phosphorus and calcium components with increasing immersion time in SBF indicates the formation of hydroxyapatite. EDX, energy-dispersive x-ray analysis; XRD, x-ray diffractometer.

3.3 | Mechanical properties of the composite material

The maximum stress and elastic modulus of B/A-Si were initially measured at 7.51 ± 2.20 and 219 ± 66.9 MPa, respectively, before immersion. However, these properties showed a decreasing trend with

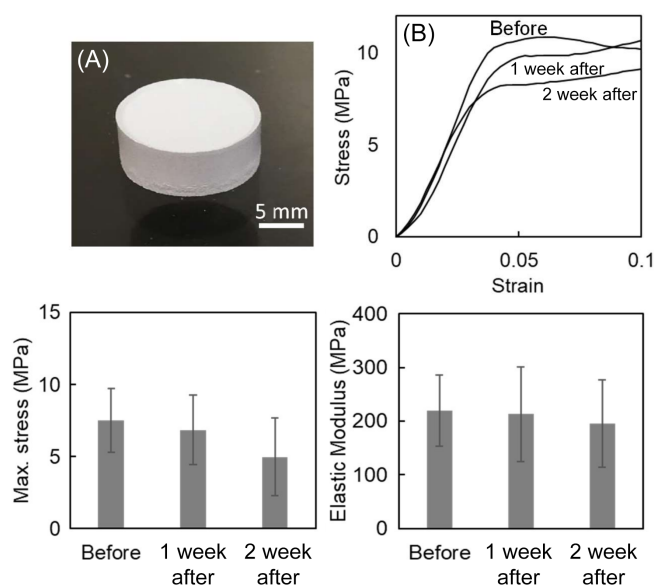


FIGURE 5 Close-up images of B/A-Si (A) and its mechanical properties: Stress-strain curve (B); maximum stress (C); elastic modulus (D) before, 1 and 2 weeks after immersion in SBF. The mechanical properties of B/A-Si show a nonsignificant but gradually decreasing trend with immersion in SBF. SBF, simulated body fluid.

increasing immersion time up to 2 weeks; nevertheless, no significant difference was found between observation days (Figure 5C,D). It is presumed that the decrease in these mechanical properties due to immersion in SBF is probably due to the leaching of calcium and swelling of alginate in this composite.

3.4 | Cytotoxicity

Regardless of the percentage of extraction fluid used, the cell viability was approximately 95%, equivalent to when fresh medium was used for cultivation, and no statistically significant differences were observed between groups (Figure 6A).

3.5 | Cellular proliferation

On each observation day, the cell numbers cultured on B/A-Si were not significantly different from those of the control group (Figure 6B). This result indicates that the composite had no effect on osteoblast proliferation.

3.6 | ALP activity

Although ALP activity values increased with increasing culture duration, they were significantly lower in the group cultured on B/A-Si than in the control group on all observation days (Figure 6C). These results indicate that the composite delays ALP activation during osteoblast differentiation.

3.7 | Cellular morphology

Apatite-like deposits were observed on the surface of the composite, which resembled those on silane cross-linked alginate and increased in size with increasing incubation time (Figure 7A–C). In contrast, the cells adhered to and actively extended on the composite (Figure 7D,E).

3.8 | Osteogenesis

After 21 days of cultivation, it was confirmed that nBMC and nBV increased significantly in the presence of the composite, but no

significant difference was observed between the presence and absence of cells (Figure 8A,B). There was no significant difference in the nBMD of the newly formed bone matrix between the groups, which was approximately 150 mg/cm³ (Figure 8C).

4 | DISCUSSION

In this study, B/A-Si, a composite of calcined bone powder and silane cross-linked alginate, was proposed as a novel bone substitute material. The biocompatibility and mechanical compatibility of B/A-Si was investigated by examining its mechanical properties, stability,

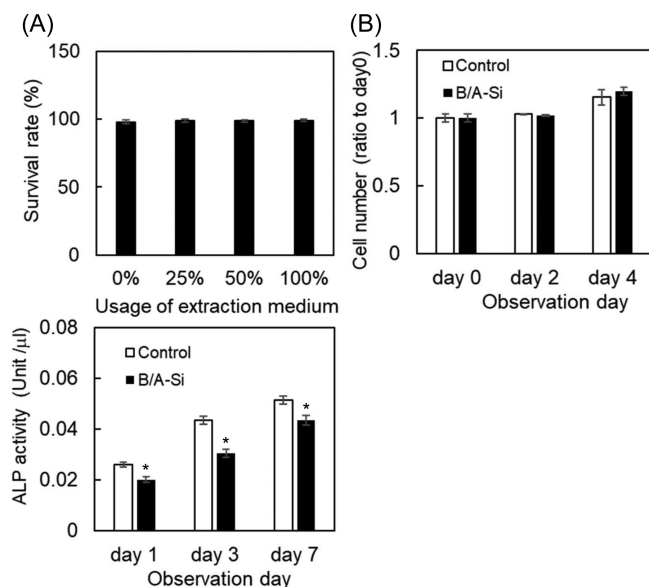


FIGURE 6 Cell affinity of B/A-Si for osteoblasts in terms of cytotoxicity (A), cell proliferation (B), and alkaline phosphatase activity (C). Symbol: * $p < .05$ versus control. B/A-Si is not cytotoxic and does not affect cell proliferation, but significantly reduces ALP activity. ALP, alkaline phosphatase.

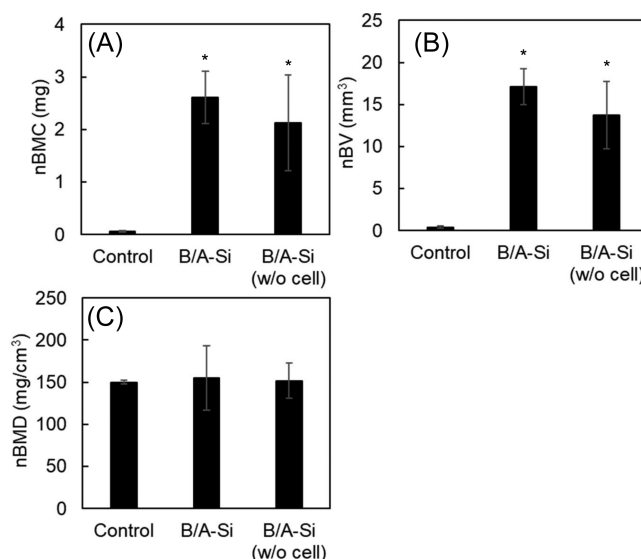


FIGURE 8 Micro-CT analysis of newly formed bone matrix on B/A-Si with or without osteoblasts: Bone mineral content (nBMC, A); bone volume (nBV, B); and bone mineral density (nBMD, C). Symbol: * $p < .05$ versus control. A new bone-like matrix is formed on the B/A-Si surface with or without cells, and there is no significant difference in the amount of new bone-like matrix between the two.

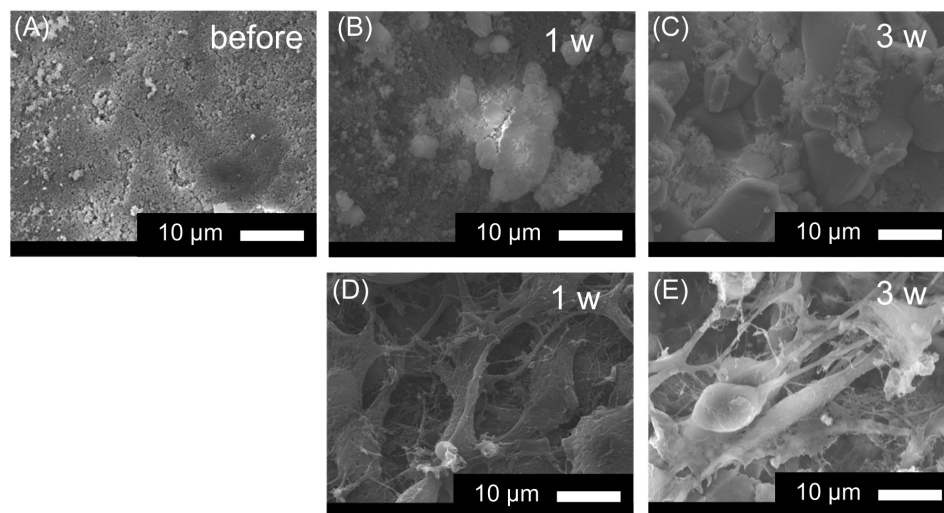


FIGURE 7 FE-SEM images of B/A-Si surfaces before (A), after 1, and 3 weeks in culture without (B, C) and with (D, E) osteoblasts. Apatite-like deposits appear on the B/A-Si surface in culture, but these also enable cell adhesion and proliferation. FE-SEM, field-emission scanning electron microscopy.

cytotoxicity, cellular affinity, and osteoconductivity. The mechanical properties of the composite were comparable to those of the human cancellous bone,²⁶ demonstrating the necessary initial rigidity and adaptability. Moreover, these properties remained relatively stable even after 2 weeks of immersion in SBF and did not deteriorate significantly. In addition, B/A-Si is not toxic to osteoblasts and has no noticeable effect on cell proliferation. However, this led to a decrease in ALP activity. In contrast, B/A-Si induced the formation of noncellular apatite. These findings suggest that this composite material has mechanical compatibility, biocompatibility, and osteoconductive properties, making it a promising bone substitute material.

Bones play an important role in supporting the body and enable mechanical functions that allow movement and work in tandem with muscles. Therefore, artificial bones require sufficient initial mechanical rigidity and stability, which is why they are traditionally made of metals or ceramics. However, the high stiffness of metals leads to stress shielding in the bone tissue around the implantation site, resulting in bone atrophy and loosening.²⁷ Ceramics cannot be used at the loading sites because of their brittleness. In the past, attempts have been made to improve the toughness of sintered hydroxyapatite, a typical artificial bone ceramic, by compositing it with collagen^{28,29} and biodegradable polymers, such as polycaprolactone (PCL)³⁰ or alginate.³¹ However, it is challenging to easily mix or combine these two materials to achieve sufficient mechanical stiffness and compatibility. For the B/A-Si composite proposed in this study, the implementation of silane cross-linking with alginate resulted in toughness, initial stiffness, mechanical stability, and mechanical properties comparable to those of human cancellous bone, thereby demonstrating mechanical compatibility.

It is essential that the substrate to which the cells adhere is non-toxic. In addition, it is important that the substrate does not inhibit cell activity. Our experimental results show that B/A-Si has no significant effect on the viability or proliferation of osteoblasts. The use of materials naturally derived from living organisms is often determined by their expected compatibility with cells. Despite the use of biocompatible materials, there are cases where the reduction in particle size can result in inhibitory effects or toxicity.³² Calcined bone, a mineral component of bone, is thought to have an affinity for osteoblasts. However, there are no reports of effects on cells when the particles are micronized. In this study, the particle size of the calcined bone reached the nanoscale, which emphasizes the importance of considering its potential effects. Nevertheless, it should be noted that the particles do not come into direct contact with the cells as they are complexed with alginate. However, the particle size of the calcined bone may alter the surface roughness of the composite, potentially affecting cell activity.³³ Furthermore, our results indicate a notable reduction in ALP activity in osteoblasts exposed to B/A-Si, suggesting a possible inhibition of cell differentiation by the composite.

In artificial bone transplantation, the osteoconductive ability to rapidly integrate with the surrounding bone is necessary to improve early stability more effectively.³⁴ Our composite, B/A-Si, showed no cytotoxicity, promoted cell proliferation, and spontaneously formed apatite on the material surface in cultivation experiments. The apatite formation was attributed to the nature of the silane cross-linked

alginate. Therefore, B/S-Si is expected to have excellent affinity and bonding with the surrounding bone. However, the osteoconductive potential of B/A-Si through cellular activation may not be significant because the ALP activity of osteoblasts on the composite surface is not high, and the observed apatite formation is noncellular. Osteoconductivity and osteoinductivity are crucial properties of artificial bone. Numerous studies have investigated the properties of materials such as hydroxyapatite, β -tricalcium phosphate (β -TCP), and titanium.^{35–37}

In this study, the mechanical stability and osteoconduction properties of B/A-Si were assessed by immersion in SBF and culture media for a maximum of 3 weeks. These properties should be evaluated by transplanting the composite into experimental animals, which would allow for a longer observation period. Transplantation experiments with laboratory animals can confirm additional biocompatible properties of the material, such as immune responses, which cannot be determined in *in vitro* experiments. Fractures in elderly people with osteoporosis are often the result of falls, with impact loading playing a dominant role in their occurrence.³⁸ However, in this study, the mechanical properties of the composite were assessed by quasi-static testing, and it was necessary to investigate the mechanical properties during impacts that could result in fractures. Previous studies have investigated the mechanical properties of bone tissue under impact loading using Hopkinson bar³⁹ and Charpy impact tests.⁴⁰ It should be noted that the impact properties of materials such as biological tissue containing water and organic components may differ from the mechanical properties determined under quasi-static conditions. Considering both the internal and external environment of the body will help us to better understand the utility of our composite as a bone replacement material.

In this study, the objective was to develop a novel bone substitute material derived from natural sources with good biocompatibility and mechanical compatibility. To achieve this objective, a composite was constructed by combining calcined bone powder obtained from bovine femoral trabecular bone with silane cross-linked alginate. The experimental results showed that this composite exhibited mechanical properties comparable to those of human trabecular bone and demonstrated mechanical stability in SBFs. Furthermore, it did not exhibit cytotoxicity and did not inhibit cell proliferation, but induced spontaneous apatite formation. We concluded, based on these results, that this composite material is a promising bone substitute material with sufficient biocompatibility, mechanical compatibility, and osteoconductivity.

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CONFLICT OF INTEREST STATEMENT

No conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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